

CreditSetu: A Dual-Path Credit Scoring & Lending Marketplace for Microfinance

Prototype Stage Progress Report for UCS503P
Proj Eval 1 (MST, First Seminar)
Submitted to: Dr. Raghav B. Venkataramaiyer
Thapar Institute of Engineering and Technology

Maanya Garg, CSED* Kashvi Bansal, CSED[†] Shreesh Gupta, CSED[‡]
Shivam Raj, CSED[§]

September 18, 2026

Abstract

CreditSetu is a dual-path credit-scoring and lending-marketplace platform for microfinance in India. Its core proposition is that every borrower builds their *own* credit score from their own repayment history, while a borrower who belongs to a Self-Help Group (SHG) gets a trust-based bootstrap head start from their group's track record and an independent borrower starts from a flat base of 450 and builds purely from scratch. Scores range from 300 to 900 and are produced by an XGBoost classifier, explained feature by feature with SHAP, and turned into plain-language improvement tips, so a borrower or lender sees not just a number but why it is what it is. Lenders formally partner with specific SHGs through an approve/reject linkage rather than lending openly to anyone, and a two-directional marketplace lets borrowers request loans (broadcast or targeted) and lenders propose offers. The system is built around four separate, role-gated logins (Borrower, Lender, SHG, Admin), each authorised server-side and scoped strictly to that role's own data, and includes a live repayment flow with a demonstration payment gateway so a borrower's own actions move their own score. This report covers the requirements, architecture, data model, scoring methodology, feasibility, what has actually been built and measured on the running system, and the testing results, followed by the use case, class and activity diagrams. The scoring model reaches a holdout ROC-AUC of 0.962 and an accuracy of 0.909; 52 of 54 automated test cases pass (96.3%), with two low-severity open defects that expose no data.

Keywords: Microfinance, Credit Scoring, Cold Start, XGBoost, SHAP, Explainable AI, Self-Help Groups, Role-Based Access Control, FastAPI, React, UML, Software Testing.

*Roll No: 1024030564

[†]Roll No: 1024030563

[‡]Roll No: 1024030565

[§]Roll No: 1024031140

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1 Introduction

... background and motivation

Microfinance is one of the main routes to formal credit for low-income households in India, and much of it runs through Self-Help Groups (SHGs): small groups, often of women, who save together, meet regularly and borrow with shared responsibility. Lending decisions in this setting still depend largely on group reputation and on the judgement of field staff. A credit bureau score is of little help, because most borrowers have never held a formal loan.

This creates two problems. First, a borrower with no history cannot be scored by a model that learns only from past repayments. This is the *cold-start* problem, and it hits independent borrowers hardest, because they have no group record to fall back on. Second, where a score does exist it is usually a single number with no reason attached: a borrower cannot tell what to improve, and a lender cannot check whether the score makes sense. Structured, auditable relationships between a specific lender and a specific SHG are also usually absent – a lender either serves an entire group’s members or none of them, with no formal, revocable agreement in between.

CreditSetu is a software system built to address these problems. Each borrower gets an individual score. SHG members start with a fair, group-earned head start, while independent borrowers start lower and build their score from their own behaviour. Every score comes with an explanation. The system also formalises the relationship between lenders and SHGs and gives each stakeholder a separate, access-controlled view. Figure 1 gives an overview of the approach.

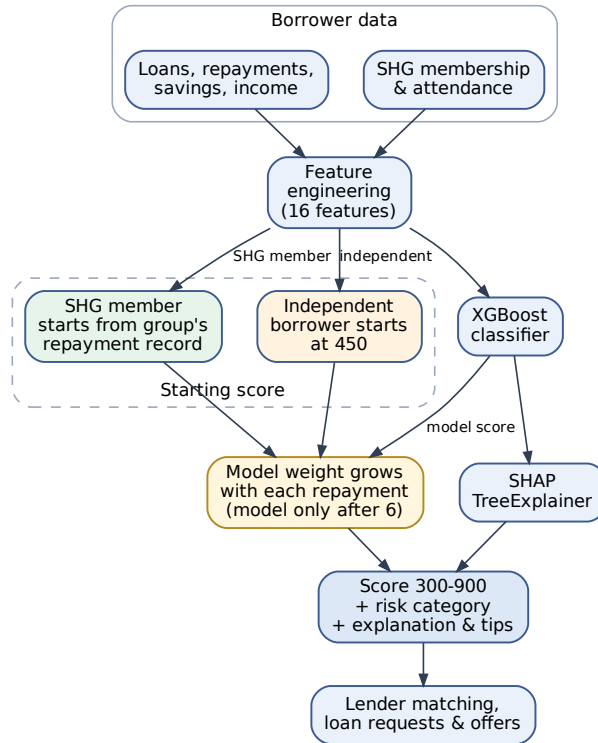


Figure 1: Overview of the dual-path scoring approach used in CreditSetu

An earlier version of this project (**Sahara MFI**, a Node/Express and MySQL application) addressed SHG-only scoring through a hand-written SQL formula, with no independent-borrower

path, no machine-learning component and no lender marketplace. CreditSetu is a from-scratch rebuild: the general domain (SHGs, loans, repayments, credit scores) carries over, but the schema, scoring approach and system design are new, driven by the dual-path requirement that did not fit the old single-path schema.

1.1 Problem Statement

- **Cold-start scoring:** an independent borrower with no formal credit history cannot be scored by a model trained only on borrowers who already have repayment history.
- **Opaque scoring:** even where a score exists, it is rarely explained in a way a borrower or a lender can act on.
- **No formal SHG–lender relationship:** lending decisions at group level are informal, unauditable and cannot be revoked cleanly.
- **Undifferentiated access:** in an open system a borrower, a lender and an SHG all see the same data, with no role-appropriate, access-controlled view for any of them.

1.2 Objectives

- Build a dual-path scoring engine that gives SHG members a fair, group-earned head start while letting independent borrowers build a score purely from their own behaviour.
- Make every score explainable, feature by feature, using SHAP, and convert the negative contributions into improvement tips.
- Formalise the SHG–lender relationship as an explicit approve/reject linkage.
- Separate the system into four role-gated views (Borrower, Lender, SHG, Admin), each authorised server-side, so no role can see another’s data.
- Let a borrower’s own actions – not just offline synthetic history – move their own score, through a live repayment flow.
- Validate the system with an automated test suite covering functionality, access control and performance.

1.3 Motivation

Groups that repay reliably have already shown the kind of discipline a lender cares about. The idea behind CreditSetu is to let that record give each member a fair starting point, and then let each member’s own repayments take over. Borrowers without a group should still be able to build a score from nothing, and both kinds of borrower should be told what is affecting their score. The project was also chosen because it meets the course selection criteria: it can be delivered in weekly increments, it is locally relevant, and it can be built entirely with mature open-source tools.

1.4 Organisation of the Report

Section 2 reviews existing approaches. Section 3 gives the requirements analysis. Section 4 describes the proposed methodology: architecture, data model, scoring, access control and the repayment flow. Section 5 is the feasibility report. Section 6 reports the implementation and the results measured on the running system, and Section 7 covers testing and validation. Section 8 concludes. The use case diagram with its written scenarios, the class diagram and the activity diagrams are given in Appendices A, B and C.

2 Related Work

... existing approaches

Traditional microfinance credit assessment relies heavily on group liability and manual field-agent judgement, with limited use of statistical or machine-learning scoring. Where formal credit

bureaus exist (for example CIBIL in India), they score individuals who already have a formal credit history and have no mechanism for a first-time, group-affiliated borrower to inherit a trust signal from their group. Commercial microfinance information systems – including the team’s own earlier **Sahara** MFI build – typically implement a single, SHG-only scoring formula with no cold-start path for independent borrowers and no per-feature explanation of the resulting score.

Gradient-boosted decision trees, and XGBoost [1] in particular, are a standard choice for tabular credit-risk data: they cope with missing values natively, need little feature scaling and train in seconds on datasets of this size. Explainable-AI techniques such as SHAP [2, 3] have seen adoption in consumer credit scoring in developed markets, where regulation increasingly demands a reason for an adverse decision, but they are far less common in microfinance-specific tooling. Isolation Forest [4] and k -means [5] are established methods for outlier detection and for grouping regions by behaviour, and are used here for anomaly flags and district-level trust clustering respectively.

What is missing in the existing work is the combination. Model opacity and the cold-start problem are usually treated as separate, unrelated concerns. CreditSetu addresses them together through an explicit bootstrap-to-model blend whose weight is driven by how much of the borrower’s own history has accumulated, and through an explanation that covers the bootstrap component as well as the model component.

3 Requirements Analysis

... what the system must do

3.1 Stakeholders

Table 1 lists the four user roles. Each role has its own login and its own dashboard.

Table 1: User roles and their main responsibilities

Role	Responsibilities in CreditSetu
Borrower	View own score, history, explanation and tips; request loans; accept or reject offers; repay EMIs.
Self-Help Group	View group statistics and member scores; request or approve partnerships with lenders.
Lender	View reputation score; browse SHGs; manage partnerships; see eligible borrowers; make offers; approve or decline loan requests.
Admin (Bank)	Read-only oversight: portfolio summary, district clusters, anomaly flags, model health and borrower search.

3.2 Functional Requirements

Table 2: Functional requirements

ID	Requirement
FR1	Provide separate logins for Borrower, Lender, SHG and Admin, and enforce role and ownership checks on the server (401 or 403).
FR2	Compute a 300–900 score for every borrower from 16 engineered features, using the dual-path starting score when there is little or no history.
FR3	Store every score as a new history row together with its SHAP explanation, and show up to five improvement tips.
FR4	Let a borrower create, retarget and withdraw loan requests; a targeted request is visible only to its target lender.
FR5	Let a lender approve (with an interest rate) or decline a request; approval creates an active loan, and a decline does not close the request for other lenders.
FR6	Let a lender propose offers to eligible borrowers within its maximum loan amount; the borrower accepts or rejects.
FR7	Manage SHG–lender partnerships; only the party that did not start the request may decide it.
FR8	Calculate the EMI and total dues, block payment if the wallet cannot cover all dues, and record the repayment and the rescore in one transaction.
FR9	Give the admin a portfolio summary, district clusters, anomaly detection, model health and borrower search.

3.3 Non-Functional Requirements

Table 3: Non-functional requirements and target values

ID	Category	Requirement
NFR1	Performance	Median response time of read endpoints below 300 ms on the demonstration dataset.
NFR2	Security	100% of cross-role or cross-identity requests refused; passwords stored only as salted hashes.
NFR3	Accuracy	Scoring model holdout ROC-AUC at least 0.85 and accuracy at least 0.80.
NFR4	Reliability	A payment and its rescore are atomic: a failed request leaves no partial change.
NFR5	Scalability	Switch from SQLite to PostgreSQL through DATABASE_URL only; all foreign keys indexed; list endpoints paginated.
NFR6	Usability	Plain-language explanations; text-size, high-contrast and dark-mode controls.
NFR7	Privacy	Only synthetic data; the public home page shows aggregate highlights only.

3.4 Use Case Model

The system has 17 use cases across the four actors. They are shown as a use case diagram in Appendix A, Figure 6, together with written scenarios (pre-conditions, main and alternate flows, exceptions and post-conditions) for the six most important of them.

4 Proposed Methodology

... *system design*

4.1 Technology Stack

Table 4 lists the tools used to build the system. Every one of them is mature, open-source and free of licensing cost.

Table 4: Technology stack

Layer	Technology
Frontend	React 19 single-page application, Vite, React Router, Recharts
Backend	Python 3.11, FastAPI (REST API with automatic OpenAPI docs), Uvicorn
Database	SQLAlchemy ORM over SQLite (demonstration); PostgreSQL through a configuration setting
Machine learning	XGBoost, SHAP, scikit-learn (k -means, Isolation Forest), pandas, NumPy
Data	Synthetic borrower population generated with Faker
Testing	pytest, FastAPI TestClient, Playwright (browser tests)
Tools	Git and GitHub, GitHub Actions (documentation site), PlantUML and Graphviz (diagrams)

4.2 System Architecture

CreditSetu is a three-tier web application:

- **Frontend:** a React (Vite) single-page application with four role-gated dashboards (Borrower, Lender, SHG, Admin) and a public landing page, styled as a government-portal-style demonstration platform (clearly labelled as such, and not impersonating any real government property).
- **Backend API:** FastAPI, exposing REST endpoints under `/api` through twelve routers, with bearer-token session authentication per role and server-side authorisation scoping every query to the authenticated caller’s own identity. Routers delegate to service modules, which are the only place business rules live.
- **Database:** SQLAlchemy over SQLite for the demonstration deployment. A one-line `DATABASE_URL` environment-variable change moves this to PostgreSQL for concurrent production use, with no code changes.
- **ML pipeline:** offline scripts (not imported by the API at request time) for synthetic data generation, model training and batch scoring, producing artifacts that the API loads once at start-up and then serves.

4.3 Data Model

The schema has 19 tables. The core of it is `districts`, `shgs` and `individuals`, where `individuals.shg_id` is the dual-path pivot: it is nullable, and NULL means the borrower is independent. Around this sit `shg_attendance_records`, `savings_records`, `lenders`, `shg_lender_links` (the approve/reject partnership table, carrying an `initiated_by_role` column so the interface can tell which side is entitled to decide a given link), `loans` (with a borrower-chosen `target_lender_id` on the request side), `repayment_events`, `credit_scores` (append-only score history, not a single mutable column), `score_explanations` (one row per feature contribution, in score points, per score), `loan_offers` and `loan_requests` (the two directions of the marketplace), `anomaly_flags`, and four separate `...sessions` tables, one per role. Every foreign key is indexed, and unique constraints prevent duplicate partnerships

and duplicate declines. The full class-level detail of every table, field and relationship is given in the class diagram in Appendix B.

4.4 Scoring Approach

Each borrower’s score is computed in the following steps. The same process is shown as an activity diagram in Appendix C, Figure 8.

1. **Collect data.** The system gathers the borrower’s loans, repayments, savings, income and, for SHG members, meeting attendance, and turns them into 16 features such as the on-time repayment rate, the missed rate, the average days late and savings regularity. Missing data is marked as unknown rather than silently treated as zero.
2. **Choose a starting score.** An SHG member starts from a score based mainly on how well the *other* members of the group repay (weighted 60% towards the group’s own record and 40% towards a neutral value of 500). An independent borrower starts from a flat 450.
3. **Blend in the model.** If the borrower has no repayments yet, the starting score is used as it is and the model is not called. With each repayment event, more weight moves from the starting score to the XGBoost model’s prediction. After six repayment events the model is used alone.
4. **Explain the score.** SHAP contributions are converted into score points and saved as explanation rows, together with a row for the starting score while it still carries weight. Contributions are measured against the score of an average borrower rather than against the floor of 300, so a borrower with a low score still receives a meaningful explanation instead of an almost empty one.
5. **Assign a risk category and save.** The score is kept within 300–900 and labelled Very Low risk (800 or more), Low (700–799), Medium (550–699), High (450–549) or Very High (below 450). Each run adds a new row to the borrower’s score history, so older scores are never overwritten.

Scoring runs for the whole population as a batch job, on demand when an admin asks for a recompute, and automatically after every repayment.

4.5 Role-Based Access Control

Four roles – Borrower, Lender, SHG and Admin – log in separately (phone and password for borrowers, username and password for the other three). Passwords are stored only as salted PBKDF2-HMAC-SHA256 hashes [6]. A successful login issues a random opaque session token that expires after seven days; there is no JWT and no client-side claim that the server trusts.

Every protected endpoint uses one of five FastAPI dependencies (one per role, plus “any logged-in role”). The borrower, lender or SHG that a request refers to is always derived from the session and never taken from the request body. An identifier appearing in a URL is checked against what the session is entitled to see, and the server answers 401 when there is no valid token and 403 when the caller is known but not entitled. Table 5 summarises what each role may see; this is enforced on the server, not merely hidden in the interface.

Table 5: Data access matrix (enforced on the server)

Data	Borrower	Lender	SHG	Admin
Own score, history, SHAP	Own only	–	–	All (read)
Other borrowers	Denied	Summary, linked SHGs or independents	Own members	All (read)
SHG details and members	Own SHG, no member list	Members if link is Approved	Own SHG	All (read)
Loan requests and offers	Own	Addressed to or made by them	Denied	All (read)
Partnership decisions	Denied	SHG-initiated links	Lender-initiated links	Denied
Anomalies, clusters, model health	Denied	Denied	Denied	Allowed

4.6 Loan Marketplace

A lender is *eligible* for a borrower if (i) the borrower is an SHG member and that SHG has an Approved link with the lender, or the borrower is independent and the lender serves independent borrowers, and (ii) the borrower’s latest score is at least the lender’s minimum score. A borrower may broadcast a loan request to every eligible lender or target one chosen lender, and may retarget a pending request or withdraw it. A lender approves a request with an interest rate, which creates an active loan, or declines it; a decline only removes the request from that lender’s queue and leaves it pending for the others. In the other direction, a lender may propose an offer to an eligible borrower within its maximum loan amount, and the borrower accepts or rejects it.

To reward competitive pricing, in the spirit of the Reserve Bank of India’s priority-sector and co-lending incentives [12], each lender carries a reputation score from 0 to 100, where 50 is the platform average. The score rises when the lender’s base rate and the rates it actually offers are below the platform average (the two heaviest factors), and rises further with marketplace activity and with willingness to serve independent borrowers. Borrowers see this score next to each suggested lender.

4.7 Repayment and Payment Gateway

Once a loan is active, the monthly instalment (EMI) is the principal plus simple interest over the full tenure, divided equally across the months. The amount due this cycle is derived from the count of existing repayment events on that loan rather than from elapsed calendar time, so a loan is payable as soon as it is disbursed instead of only after a real due date has passed – necessary for the feature to be usable by a live user rather than only replayable against backdated data.

A demonstration payment gateway (clearly labelled as not real) shows the borrower’s wallet balance and the amount due per active loan. Before any payment the server reloads the wallet and recomputes the borrower’s *total* dues across all active loans; if the wallet cannot cover the total, every payment is blocked, not just the one clicked, so a borrower cannot clear one loan while silently defaulting on another. A successful payment creates a real repayment event, reduces the outstanding balance, closes the loan after the final instalment, debits the wallet and recomputes the credit score – all inside a single database transaction, so a failure leaves no partial change. The full flow is shown as a swimlane activity diagram in Appendix C, Figure 9.

5 Feasibility Report

... can this actually be built and run

5.1 Technical Feasibility

Every component uses mature, well-documented open-source tooling already proven at this scale: FastAPI and SQLAlchemy for the API and the ORM, XGBoost and SHAP for scoring and explanation, scikit-learn for k -means district clustering and Isolation Forest anomaly detection, and React/Vite for the frontend. No component requires novel research or unavailable infrastructure. The dataset is small enough (1,079 borrowers, 494 training rows) that model training takes seconds on a two-core machine, and the API serves the trained artifacts rather than retraining at request time. The team has already built, run and fixed real defects in this stack (Section 6.5), which demonstrates that the architecture is technically sound in practice and not only on paper. The main technical risk – that the model would be unusable for borrowers without history – is exactly what the dual-path starting score is designed to remove, and that behaviour has been verified on the running system (Section 6.3).

5.2 Operational Feasibility

The system runs on a single machine for demonstration (SQLite, `uvicorn --reload`, `npm run dev`) with no dependency on any external paid service. Each of the four stakeholder roles has a dashboard built around the decision that role actually has to make, and the borrower view deliberately avoids jargon: the score is accompanied by a plain-language explanation and up to five concrete improvement tips, with text-size, high-contrast and dark-mode controls for low-literacy and low-vision users. A production deployment would swap SQLite for a managed PostgreSQL instance through a single environment variable and replace the deliberately minimal demonstration credential scheme with production-grade hashing and short-lived signed tokens; both are additive changes, not architectural rewrites. Field operation would additionally need an SHG facilitator to keep attendance and savings records current, which is already part of normal SHG practice.

5.3 Economic Feasibility

All tooling used is free and open-source, so there is no licensing cost. Development cost is the team’s own effort within the course timeline. At production scale the only recurring cost would be standard cloud hosting – a small managed database instance and an API/web server – which is proportionate to any comparable small-to-medium deployment and is not a blocker for a course project or an early pilot. Because the scoring model is trained offline and only the artifacts are loaded by the API, there is no per-request inference cost and no paid model-serving dependency.

5.4 Schedule Feasibility

The eight-stage plan (synthetic data generation, scoring engine, cold-start bootstrap, SHG–lender linkage, lender matching and offers, frontend, geo-clustering, anomaly detection) was completed and verified end to end before the current iteration began. The current iteration – role separation, server-side authorisation, targeted requests and the live repayment flow – is complete, tested against the live API and a real browser, and documented with before/after verification (Section 6). Table 6 shows the remaining milestones, which fit within the semester timeline because each is an extension of working code rather than new construction.

Table 6: Remaining schedule to the Final Report

Milestone	Work	Status
Prototype stage, W7 (this report)	Role separation, live repayment flow, test suite, UML design	Complete
Hardening	Fix defects D-01 and D-02; production password and token scheme	Next
Analytics depth	Periodic re-scoring so score-history trends accumulate	Next
Admin actions	Moderation beyond the current read-only oversight, with an audit trail	Planned
Final Report	Fairness sensitivity analysis of the starting-score gap; full regression run	Planned

6 Implementation and Results

... what actually works, verified

All figures in this section are measured against the running system, not estimated.

6.1 Experimental Setup and Dataset

The backend uses Python 3.11, FastAPI 0.115, SQLAlchemy 2.0, XGBoost 2.1, SHAP 0.46 and scikit-learn 1.5; the frontend uses React 19 with Vite and Recharts. All measurements were taken on a two-core Linux machine with 8 GB of RAM. Because real borrower data is sensitive, the system uses a reproducible synthetic population generated with Faker (random seed 42); its size is given in Table 7. Defaults in the synthetic data follow a smooth probabilistic rule based on missed, late and partial payments rather than a hard threshold, which was itself the fix for an early defect.

Table 7: Seeded dataset

Entity	Count	Entity	Count
Borrowers (total)	1,079	SHGs	51
SHG-linked	659	Lenders	6
Independent	420	Districts	10
Loans	1,619	SHG-lender links	97
Defaulted	315 (19.4%)	Training rows for the model	494

6.2 Scoring Model Performance

The model is trained on SHG-linked borrowers with at least three repayment events (494 rows), using an 80/20 stratified split. The hyperparameters are 200 trees, maximum depth 4, learning rate 0.05, and subsample and column-sample ratios of 0.9. Table 8 gives the holdout results and the most important features.

Table 8: Scoring model: holdout performance and top features by importance

Metric	Value	Feature	Importance
Holdout accuracy	0.909	<code>missed_rate</code>	0.239
Holdout ROC-AUC	0.962	<code>on_time_rate</code>	0.201
Training rows	494	<code>attendance_rate</code>	0.081
Test split	20% (stratified)	<code>n_closed_loans</code>	0.048
Features	16	<code>avg_days_late</code>	0.047
Borrowers scored	1,079 / 1,079	<code>n_repayment_events</code>	0.045

As expected, repayment behaviour (the missed and on-time rates) dominates, followed by SHG meeting attendance. Both targets in NFR3 are met with a wide margin. Figure 2 compares the latest scores of the two borrower groups: SHG-linked borrowers average 718.7 against 675.0 for independent borrowers. Many independent borrowers sit in the High-risk band (450–549) precisely because their score starts at 450 and has not yet been raised by repayments – the dual-path design working as intended rather than a defect.

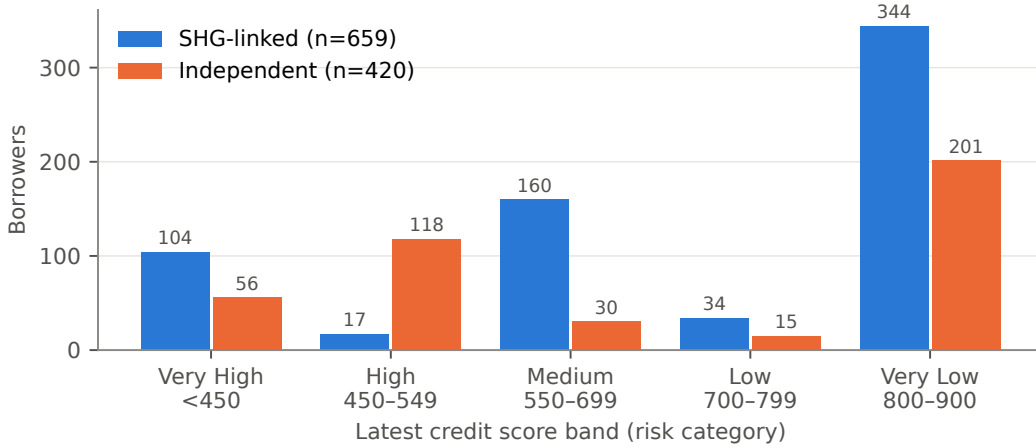


Figure 2: Distribution of latest credit scores by risk band for SHG-linked and independent borrowers

6.3 Live Validation and Performance

A seeded SHG member was taken through the full flow on the live system with no manual database edits. The borrower requested Rs. 15,000 for 12 months from District Cooperative Bank, the lender approved it at 12.5% (an EMI of Rs. 1,406.25), and the borrower paid four instalments on time. The score rose from 717 to 898 and the scoring basis switched from the blended starting score to the model alone once the borrower reached six repayment events (Figure 3a) – exactly the behaviour specified in Section 4.4.

Median response times measured on the running server are shown in Figure 3b. Login is deliberately slow (about 120 ms, because of PBKDF2 with 200,000 iterations); every read endpoint is far below the 300 ms target of NFR1, the slowest being the eligible-borrowers list at about 154 ms.

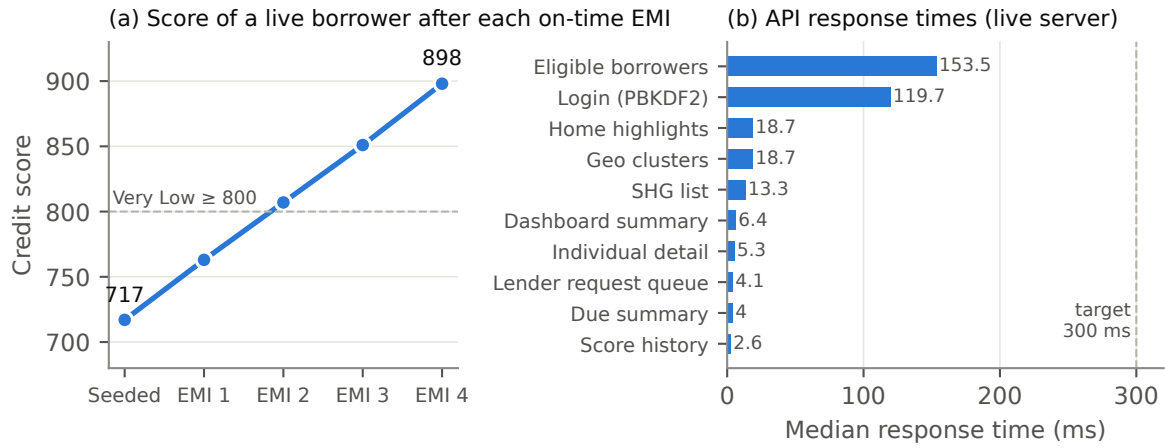
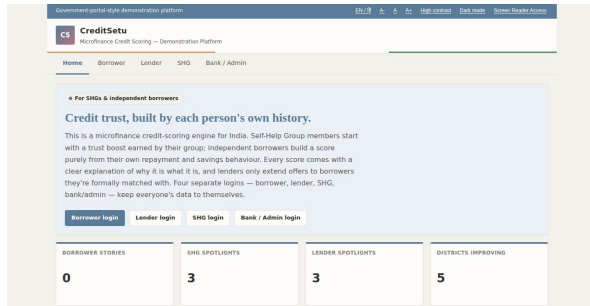


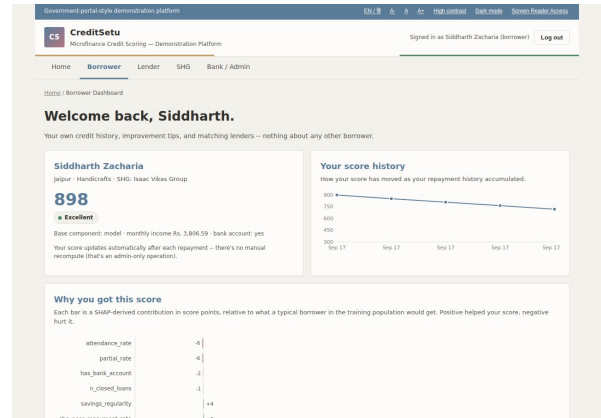
Figure 3: (a) Score of a live borrower after each on-time EMI; (b) median API response times

6.4 User Interface

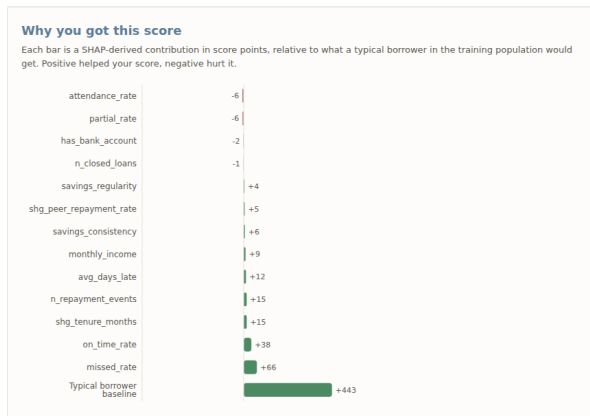
Figures 4 and 5 show screens captured from the running application.



(a) Public home page with role logins



(b) Borrower dashboard and score history



(c) “Why you got this score” (SHAP)

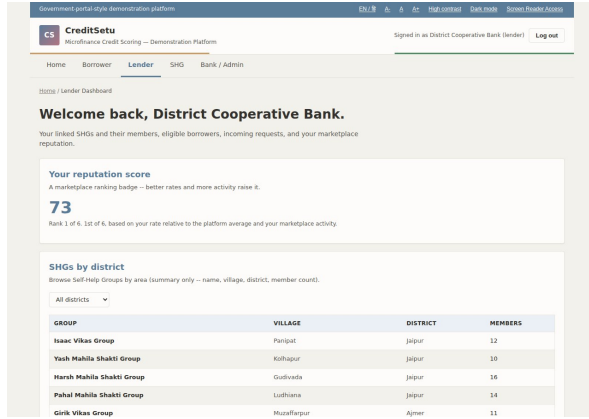
(d) Loan request (broadcast or targeted)

(e) Repay page

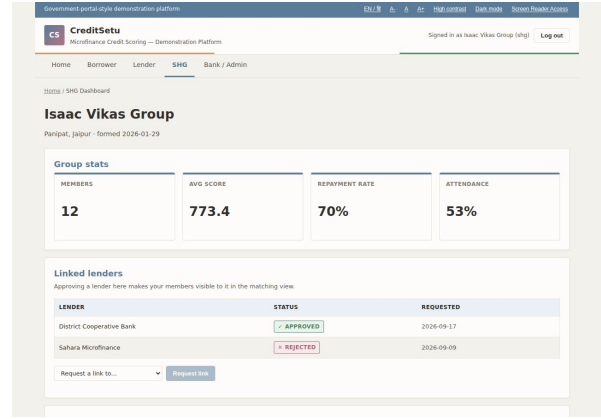
(f) After payment

(g) Payment blocked: insufficient wallet

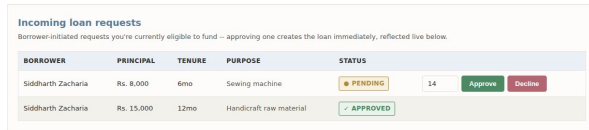
Figure 4: Borrower screens



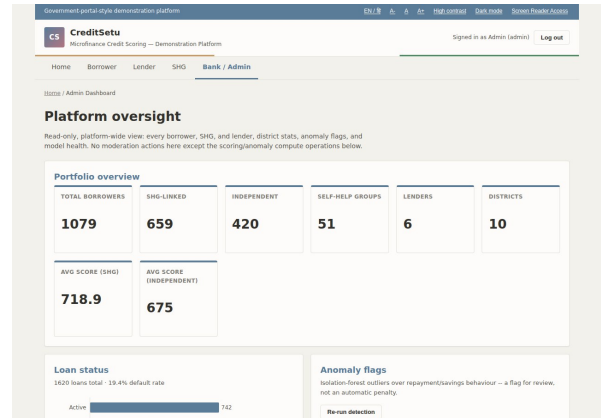
(a) Lender dashboard with reputation



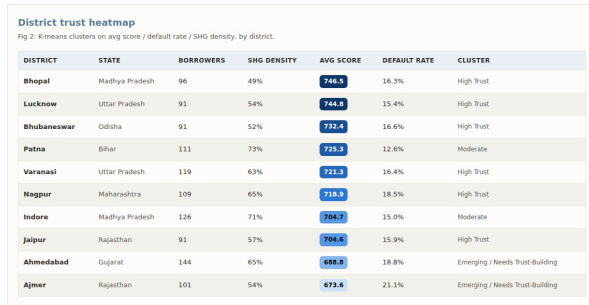
(b) SHG dashboard



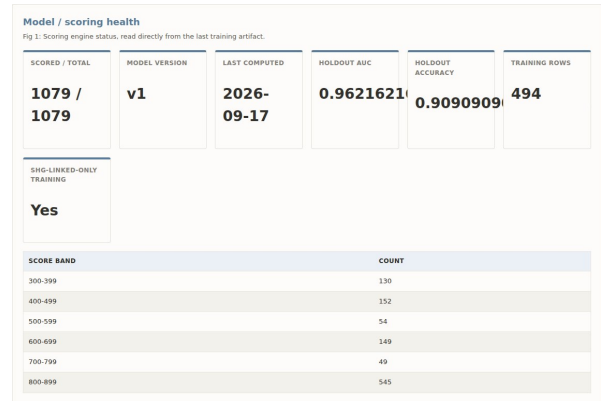
(c) Lender: incoming loan requests



(d) Admin: platform overview



(e) Admin: district trust heatmap (*k*-means)



(f) Admin: model health

Figure 5: Lender, SHG and Admin screens

6.5 Defects Found and Fixed During Development

Running the system end to end – training on the real dataset, calling the live API and exercising the live frontend – rather than only reading the code revealed several defects that shaped the design (Table 9). Two of them were silent: nothing crashed and nothing looked empty, which is precisely why they survived code review.

Table 9: Main defects found and fixed during development

Defect	Fix
A hard <code>missed_fraction > 0.4</code> cliff made the synthetic default label a near coin-flip at the boundary	Replaced with a smooth, probabilistic default rule
SHAP explanation was almost empty for high-risk borrowers, who need it most	Share points above the population baseline instead of above the floor of 300
k -means cluster labels were inverted, so the worst districts were labelled “High Trust”	Label clusters by the rank of their mean score
Approve/Reject Partnership buttons sent an undefined field (<code>link_id</code> instead of <code>id</code>), so a lender could never actually decide a link	Corrected the field name and verified the flow live
Search could reach only about half of the borrowers (interface limit of 1000 against an API cap of 500)	Search moved to the server (<code>?search=</code>) with a debounced input
GET <code>/api/shgs</code> used an N+1 query walking members, loans and repayment events per SHG (1.1–1.3 s)	Three GROUP BY aggregates: about 13 ms, a 75× improvement with identical content
Foreign keys were not indexed, and the eligible-borrower table rendered 351 rows uncapped	Indexes added; table capped at 25 rows with a “top N of M” note

7 Testing and Validation

... does it do what it claims

7.1 Test Strategy

The goal of testing is to confirm that each role can do exactly what it is allowed to do and nothing more, that scores are correct at the boundaries, and that money-related actions are atomic. Test cases were designed with equivalence partitioning (valid, invalid and unauthorised inputs) and boundary value analysis (risk bands and the ends of the score range), then automated at the three levels shown in Table 10. API tests run against a temporary copy of the seeded database, so the demonstration data is never modified, and the borrower/lender pairs used by a test are chosen at run time from the current eligibility rules instead of being hard-coded, so the suite does not silently rot as the data changes.

Table 10: Test levels and tools

Level	What is tested	Tool	Cases
Unit	Scoring formulas, risk boundaries, reputation logic	pytest	8
Integration (API)	REST endpoints with real tokens: authentication, access control, marketplace, payments, links, offers, analytics	pytest + FastAPI TestClient	38
System (end-to-end)	User journeys in a real browser	Playwright (Chromium)	8
Total			54

7.2 Test Cases and Results

Table 11 gives the step-by-step form of one representative test case, and Table 12 lists all 54 test cases with their expected and actual results from the run of 17 September 2026. Parameterised cases (TC04 over 3 roles and TC21 over 9 boundary scores) run several times, so there are 64 executions in total; the whole suite completes in 28 s.

Table 11: Detailed test case: EMI repayment through the payment gateway

Test case: TC32–TC35, E2E05–06 Subsystem: Payments Executed: 17/09/2026				
Pre-condition: borrower has an active loan of Rs. 6,000 at 12% for 6 months (EMI = Rs. 1,060)				
Step	Action	Expected	Actual	Status
1	Open due summary	Total = sum of EMIs; sufficient if wallet $\geq$ total	Consistent	Pass
2	Pay with wallet Rs. 1,00,000	Rs. 1,060 paid, On_Time, balance 4,940, wallet 98,940, new score	As expected	Pass
3	Pay with wallet Rs. 1	400 insufficient_funds , no change	Outstanding still 6,000	Pass
4	Another borrower pays this loan	403	403	Pass
5	UI: click <i>Pay this loan</i>	“New score” shown	“898 · Excellent”	Pass
6	UI: wallet of Rs. 5	Every payment blocked, shortfall shown	Banner shown	Pass
Post-condition: repayment, balance, wallet and score change together, or not at all.				

Table 12: Test cases and results

ID	Module	Test case	Expected	Actual	Status
TC01	Auth	Borrower login with valid phone and password	200 + token	200, token issued	Pass
TC02	Auth	Borrower login with wrong password	401 “Incorrect...”	401	Pass
TC03	Auth	Login with an unregistered phone number	401	401	Pass
TC04	Auth	Lender, SHG and admin login (3 runs)	200 each	200 each	Pass
TC05	Auth	Admin login with wrong password	401	401	Pass
TC06	Auth	Protected endpoints called without a token	401	401	Pass
TC07	Auth	Reuse a token after logout	401	401	Pass
TC08	Auth	Each account has a uniquely salted hash	Different hashes	All 1,079 borrowers share one hash	Fail
TC09	RBAC	Borrower reads another borrower’s record	403	403	Pass
TC10	RBAC	Borrower calls admin-only borrower list	403	401 (still denied)	Fail
TC11	RBAC	Admin lists the population	200	200	Pass

ID	Module	Test case	Expected	Actual	Status
TC12	RBAC	Lender opens an unlinked SHG	403	403	Pass
TC13	RBAC	Lender opens an approved-linked SHG	200 with members	200 with members	Pass
TC14	RBAC	Borrower opens own SHG	Summary, no members	As expected	Pass
TC15	RBAC	Anomaly flags for borrower vs admin	Denied / 200	Denied / 200	Pass
TC16	RBAC	Public home highlights without login	200	200	Pass
TC17	Scoring	New independent borrower (0 repayments)	Exactly 450	450	Pass
TC18	Scoring	New SHG member (0 repayments)	Group-based start, above 450	As expected	Pass
TC19	Scoring	All stored scores within range	300–900	In range	Pass
TC20	Scoring	Explanation rows add up to score – 300	Within tolerance	Within tolerance	Pass
TC21	Scoring	Risk band at 9 boundary scores	Correct band	9/9 correct	Pass
TC22	Scoring	Probability-to-score conversion	Scores 300, 600, 900 at the ends and middle	As expected	Pass
TC23	Scoring	Model quality targets	AUC ≥ 0.85 , acc. ≥ 0.80	0.962 / 0.909	Pass
TC24	Requests	Principal = 0	400	400	Pass
TC25	Requests	Target an ineligible lender	400	400	Pass
TC26	Requests	Targeted request visibility	Only target lender	Only target lender	Pass
TC27	Requests	Lender declines a broadcast request	Still Pending	Pending, hidden from that lender	Pass
TC28	Requests	Approve without an interest rate	400	400	Pass
TC29	Requests	Approve with rate 12%	Active loan created	Loan Active, balance 6,000	Pass
TC30	Requests	Withdraw the same request twice	Withdrawn, then 400	As expected	Pass
TC31	Requests	Withdraw another borrower's request	403	403	Pass
TC32	Payments	Due-summary totals	Sum of EMIs	Consistent	Pass
TC33	Payments	Pay EMI (Rs. 6,000, 12%, 6 months)	Rs. 1,060 paid, On_Time, new score	Exactly as expected	Pass
TC34	Payments	Wallet of Rs. 1	400, nothing changed	400, balance unchanged	Pass
TC35	Payments	Pay another borrower's loan	403	403	Pass
TC36	Links	SHG requests; SHG self-approves; lender approves	Pending; 403; Approved	As expected	Pass

ID	Module	Test case	Expected	Actual	Status
TC37	Links	Duplicate partnership request	400	400	Pass
TC38	Links	Re-decide an approved link	400	400	Pass
TC39	Links	Admin tries to decide a link	403	403	Pass
TC40	Offers	Offer to an ineligible borrower	400	400	Pass
TC41	Offers	Offer above the maximum loan amount	400	400	Pass
TC42	Offers	Wrong borrower, then right borrower, accepts	403; loan created	As expected	Pass
TC43	Analytics	Lower base rate increases reputation	Higher score	Higher score	Pass
TC44	Analytics	Best district is labelled High Trust	Correct label	Correct	Pass
TC45	Analytics	Dashboard counts add up	Consistent	Consistent	Pass
TC46	Performance	Median time of GET /api/shgs	< 300 ms	≈13 ms	Pass
E2E01	UI	Borrower login opens own dashboard	Welcome + SHAP card	Shown	Pass
E2E02	UI	Wrong password message	“Incorrect credentials.”	Shown	Pass
E2E03	UI	Open /admin without a session	Redirect to login	Redirected	Pass
E2E04	UI	Borrower session opens /lender	Lender data not shown	Not shown	Pass
E2E05	UI	Pay an EMI from the Repay page	“New score” shown	Shown	Pass
E2E06	UI	Repay with a wallet of Rs. 5	Blocked banner	Shown	Pass
E2E07	UI	Lender dashboard content	Reputation + requests	Shown	Pass
E2E08	UI	Log out, then open /shg	Redirect to login	Redirected	Pass

7.3 Test Summary

Table 13 summarises the results by module: 52 of 54 test cases passed (96.3%).

Table 13: Test summary by module

Module	Cases	Passed	Failed	Pass rate	Observation
Authentication	11	10	1	90.9%	Shared seeded hash (D-01)
Access control	10	9	1	90%	401 instead of 403 (D-02); no data leaked
Scoring engine	7	7	0	100%	Starting scores exact; AUC 0.962
Loan requests	8	8	0	100%	Rules enforced on the server
Payments	6	6	0	100%	Atomic; blocked on low funds
Partnerships and offers	7	7	0	100%	Only the correct party can decide
Analytics and performance	5	5	0	100%	Median ≈ 13 ms
Total	54	52	2	96.3%	

7.4 Defect Report

Table 14: Open defects found by testing

ID	Description	Severity	Recommended fix
D-01	<code>data_gen.py</code> hashes the demonstration password once and reuses the result, so all 1,079 seeded borrowers share one salt and hash. Lender and SHG hashes are unique.	Low (demo data)	Call <code>hash_password()</code> separately for each borrower.
D-02	Admin-only endpoints answer 401 “Not logged in” to a valid borrower, lender or SHG token instead of 403 as the specification requires. Access is still denied.	Low	Return 403 in <code>get_current_admin</code> when the token belongs to another role’s valid session.

Both defects are low severity and neither exposes any data: D-01 concerns seeded demonstration accounts only, and D-02 refuses the request correctly but with the wrong status code, which misleads the frontend rather than the security check. Every other functional requirement and every measurable target (NFR1–NFR4) was met.

8 Conclusion and Future Scope

... where this stands and what’s next

At the prototype stage CreditSetu has a working, explainable, dual-path scoring engine; a formal SHG–lender linkage; a two-directional loan marketplace with targeted requests; four separately authorised role-based views with server-side access control; and a live borrower repayment flow whose effect on the credit score has been verified on a real account rather than merely claimed, with a score rising from 717 to 898 across four on-time instalments. The XGBoost model reaches a holdout ROC-AUC of 0.962 and an accuracy of 0.909, every score is explained with SHAP and turned into improvement tips, and 52 of 54 automated test cases pass. Real defects, including one that silently broke a core lender action and one with a measured $75\times$ performance impact, were found and fixed by exercising the live system rather than only reading the code.

Remaining work towards the Final Report: fixing defects D-01 and D-02; replacing the deliberately minimal demonstration credential scheme with a vetted password-hashing library and short-lived signed tokens; periodic re-scoring so that genuine score-history trends accumulate

over time (currently limited by how few times the batch-scoring pipeline has been run); admin moderation actions beyond the present read-only oversight, with an audit trail; integration with a real payment provider and EMI reminders; retraining on real data with a time-based train/test split; and a mobile, multilingual interface for rural borrowers. The gap between the independent and SHG starting scores remains a stated policy parameter rather than something derived from the data, and is flagged as a candidate for a fairness sensitivity analysis in the Final Report.

Data and Code Availability

The source code, documentation and synthetic data generator are available at <https://github.com/shreesh1802/ucs503p-202627-Microfinance-Project>.

A Use Case Diagram

Figure 6 shows the four actors (Borrower, Self-Help Group, Lender and Admin) against every use case, each of which is grounded in an API router and a dashboard that exist in the code. Log In / Log Out and the lender directory are shared by several actors; the remaining use cases belong to a single role. Two *include* relationships are worth noting: viewing the score always includes generating the improvement tips from it, and repaying an EMI always includes recomputing the credit score, which is why a repayment cannot leave the score stale.

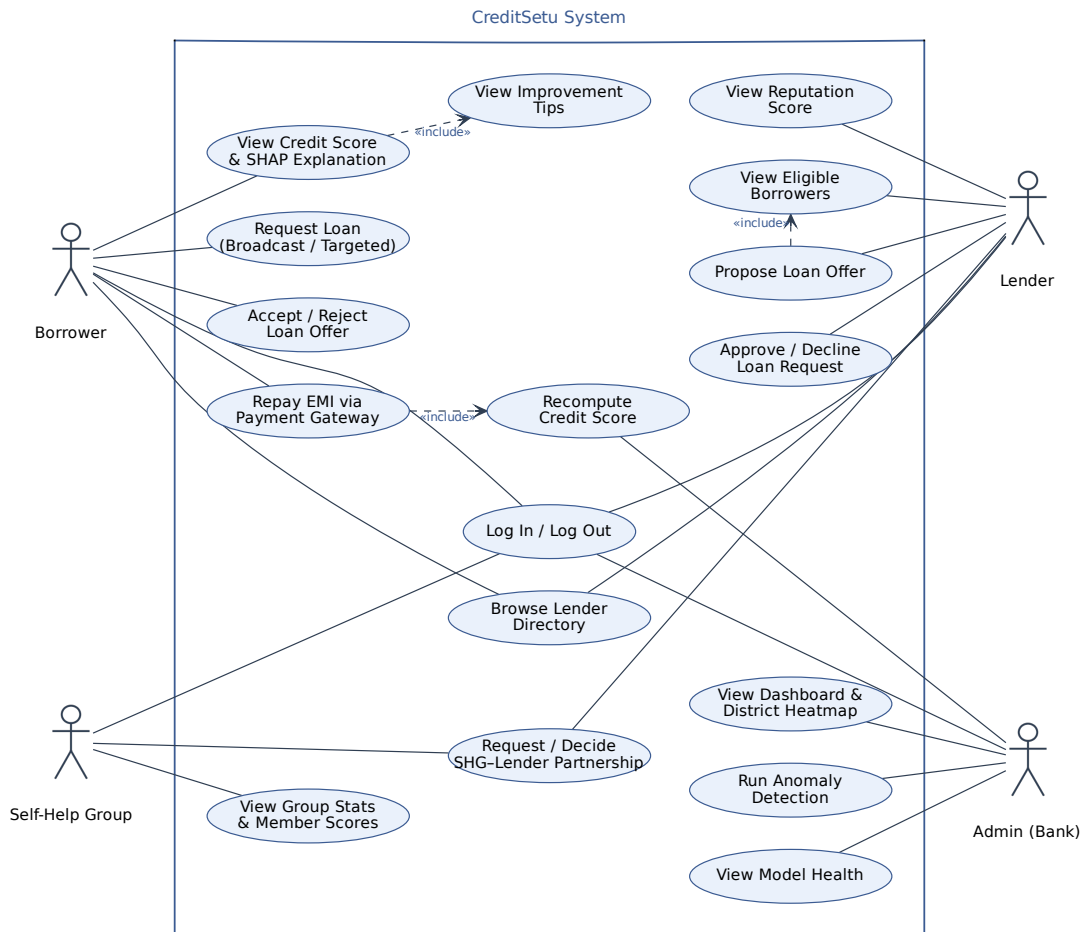


Figure 6: Use case diagram of CreditSetu

Use Case Scenarios

Tables 15–20 give the written scenarios for the six most important use cases. HTTP status codes in the exception rows are the codes the API actually returns.

Table 15: Use case specification: Log In

Use case	UC01: Log In (role-gated)
Actors	Borrower, Self-Help Group, Lender, Admin
Pre-conditions	The account exists in the seeded demonstration data; the user has opened the login page for their role.
Main flow	(1) User selects a role and submits credentials: phone and password for a borrower, username and password for the other three roles. (2) System looks up the account for that role only. (3) System verifies the password against the stored salted PBKDF2 hash. (4) System creates a row in that role’s session table and returns an opaque bearer token valid for seven days. (5) The frontend stores the token and opens that role’s dashboard.
Alternate flows	The user logs out, which deletes the session row; any later call with the same token is refused.
Exceptions	Unknown identifier or wrong password: HTTP 401 with the message “Incorrect credentials.” and no indication of which of the two was wrong. A token used against another role’s endpoints is refused.
Post-conditions	A session row exists for exactly one role, and the caller’s identity for every later request is taken from that session rather than from the request body.

Table 16: Use case specification: View Credit Score and Explanation

Use case	UC02: View Credit Score, Explanation and Improvement Tips
Actor	Borrower (Admin may read any borrower’s score)
Pre-conditions	Borrower is logged in and has at least one stored score.
Main flow	(1) System takes the borrower’s id from the session. (2) System returns the latest score, its risk band and the scoring basis (starting score, blended or model). (3) System returns the score history so the dashboard can draw the trend. (4) System returns the stored per-feature contributions in score points. (5) The dashboard shows them as a bar chart and converts the negative contributions into up to five plain-language improvement tips.
Alternate flows	An admin may trigger a recompute, which appends a new score row rather than overwriting the existing one.
Exceptions	Requesting another borrower’s score: HTTP 403. No token: HTTP 401.
Post-conditions	Nothing is modified; the score history is read-only from this use case.

Table 17: Use case specification: Request a Loan

Use case	UC03: Request a Loan (broadcast or targeted)
Actor	Borrower
Pre-conditions	Borrower is logged in and has a credit score.
Main flow	(1) Borrower enters principal, tenure and an optional purpose. (2) Borrower chooses <i>broadcast</i> or one specific eligible lender. (3) System takes the borrower’s id from the session and checks that the principal and tenure are positive. (4) For a targeted request, system checks that the lender is still eligible: an approved SHG link or willingness to serve independents, and a score at or above the lender’s minimum. (5) System saves the request as <i>Pending</i> and shows it under “Your loan requests”.
Alternate flows	Borrower retargets a pending request to another lender or back to broadcast, or withdraws it.
Exceptions	Invalid amount or ineligible target: HTTP 400 with a message. Acting on another borrower’s request: HTTP 403. Withdrawing an already withdrawn request: HTTP 400.
Post-conditions	The request appears in the queue of every eligible lender, or only in the target lender’s queue.

Table 18: Use case specification: Approve or Decline a Loan Request

Use case	UC04: Approve or Decline a Loan Request
Actor	Lender
Pre-conditions	Lender is logged in; at least one pending request is addressed to it, either directly or by broadcast.
Main flow	(1) Lender opens the incoming request queue, which contains only requests it is eligible for. (2) Lender opens a request and sees the borrower’s score, risk band and path (SHG-linked or independent). (3) To approve, the lender enters an interest rate. (4) System re-checks eligibility and the lender’s maximum loan amount at the moment of the decision, not at the moment of the request. (5) System marks the request approved and creates an Active loan with its outstanding balance and EMI schedule.
Alternate flows	The lender declines: the request is hidden from that lender but remains <i>Pending</i> for every other eligible lender, and the same lender cannot decline it twice.
Exceptions	Approval without an interest rate: HTTP 400. Deciding a request addressed to another lender, or an admin attempting to decide: HTTP 403. Re-deciding a request that is no longer pending: HTTP 400.
Post-conditions	Either an Active loan exists and the request is closed, or the request remains open for other lenders. No intermediate state is stored.

Table 19: Use case specification: Repay EMI through the Payment Gateway

Use case	UC05: Repay EMI
Actor	Borrower
Pre-conditions	Borrower is logged in, has at least one active loan and has signed in to the demonstration gateway.
Main flow	(1) System shows the wallet balance, the EMI for each active loan and the total due this cycle. (2) Borrower clicks <i>Pay this loan</i> . (3) System reloads the wallet and recalculates the total due on the server, ignoring the values the page was showing. (4) System records the repayment as <i>On-Time</i> or <i>Late</i> , debits the EMI from the wallet, reduces the outstanding balance and closes the loan after the final instalment. (5) System recomputes the credit score and commits everything in one transaction. (6) The page shows the new score.
Alternate flows	A late payment is recorded with the number of days late, which feeds the <code>avg_days_late</code> feature at the next scoring run.
Exceptions	Wallet below the <i>total</i> dues across all active loans: HTTP 400 <code>insufficient_funds</code> , and every payment is blocked, not only the one clicked. Paying another borrower's loan: HTTP 403.
Post-conditions	Loan, wallet and score history are updated together, or not at all.

Table 20: Use case specification: Request and Decide an SHG–Lender Partnership

Use case	UC06: Request / Decide an SHG–Lender Partnership
Actors	Self-Help Group, Lender
Pre-conditions	Both parties exist; no approved or pending link already joins them.
Main flow	(1) Either party browses the other (an SHG browses the lender directory, a lender browses SHGs by district) and sends a partnership request. (2) System stores the link as <i>Pending</i> together with the role that initiated it. (3) The other party sees it in its pending list and approves or rejects it. (4) On approval the link becomes <i>Approved</i> , and from that moment the lender may see that SHG's member list and its members may target that lender.
Alternate flows	Either party may later revoke an approved partnership, after which member visibility is withdrawn.
Exceptions	The initiating side deciding its own request: HTTP 403 (only the other party may decide). A duplicate request, or a decision on an already-decided link: HTTP 400. An admin attempting to decide: HTTP 403, because that decision belongs to the two named parties.
Post-conditions	Exactly one link row exists for the pair, in state Pending, Approved or Rejected, with a record of which side initiated it.

B Class Diagram

Figure 7 shows the domain model, which matches the SQLAlchemy schema in `app/models.py`. Three design decisions are visible in it:

- `Individual.shg_id` is optional, and a missing value marks an independent borrower. This single nullable association carries the whole dual-path design without duplicating tables or splitting the population into two parallel schemas.
- `CreditScore` is append-only and carries its own `ScoreExplanation` rows, so any past score and the reason for it can be reconstructed later. Nothing overwrites a score in place.
- `SHGLenderLink` is a many-to-many association table that also records its own state and which side initiated it, which is what makes the approve/reject rule enforceable.

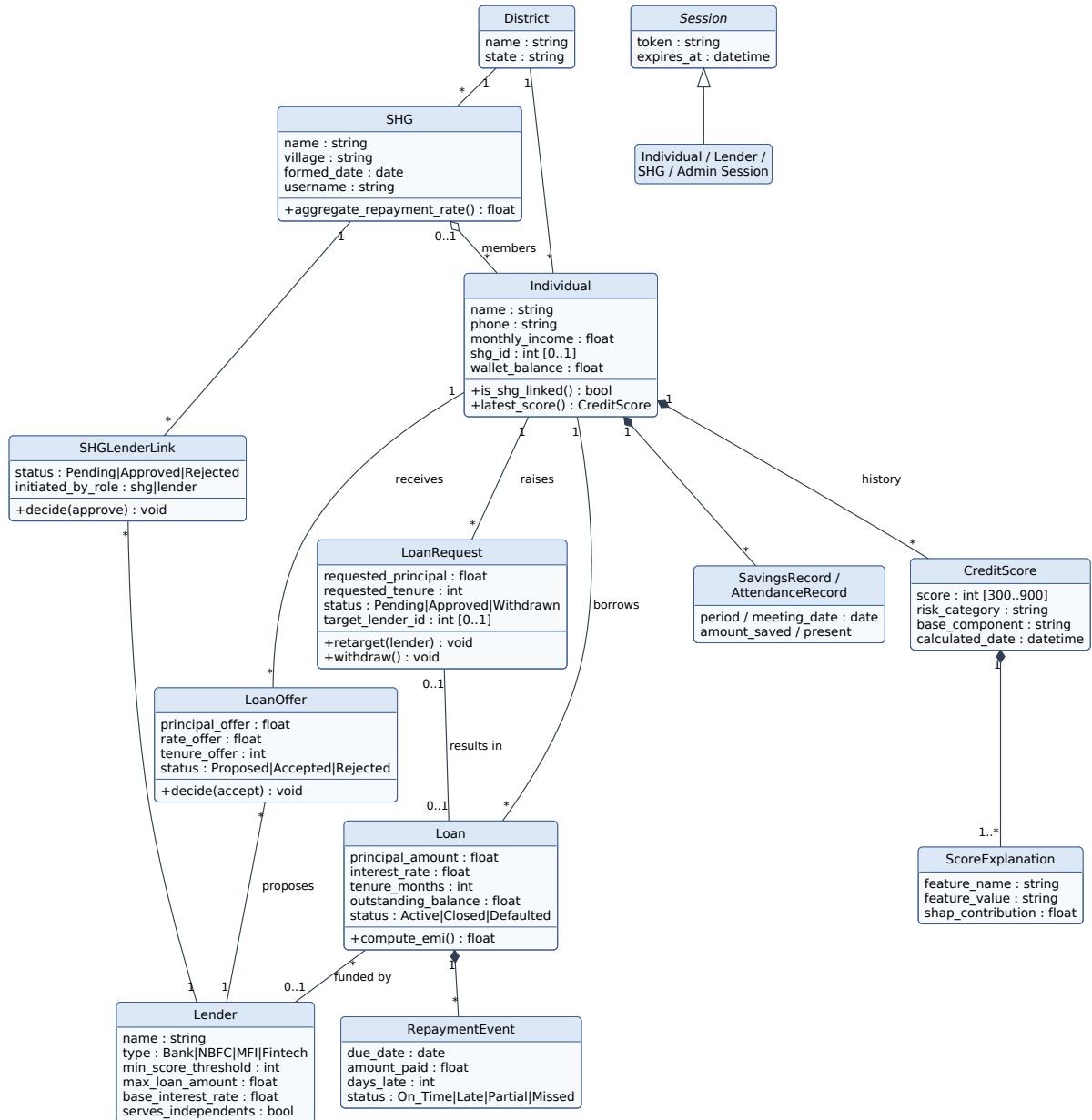


Figure 7: Class diagram (domain model)

C Activity Diagram

Figure 8 shows the credit-scoring activity described in Section 4.4: the branch on SHG membership selects the starting score, and the branch on repayment history decides whether the model is called at all, which is how the cold-start case is handled without a separate code path.

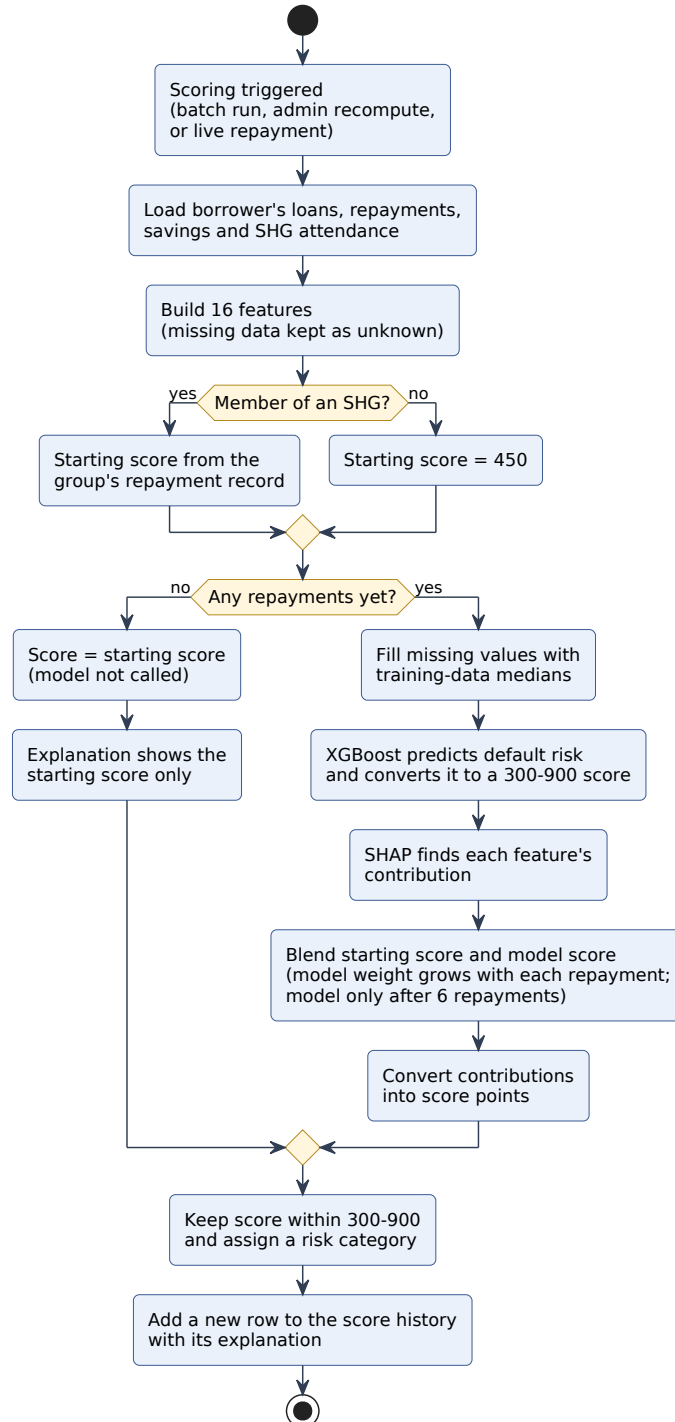


Figure 8: Activity diagram: computing a borrower's credit score

Figure 9 shows the same notation applied to the main business process, with one swimlane per participant: the borrower's loan request, the lender's decision, and the repayment that updates the loan, the wallet and the score in a single transaction. The two stop nodes on the left of the diagram are the guards described in Section 4: an invalid or ineligible request never reaches a lender's queue, and a wallet that cannot cover the borrower's total dues blocks every payment rather than only the one clicked.

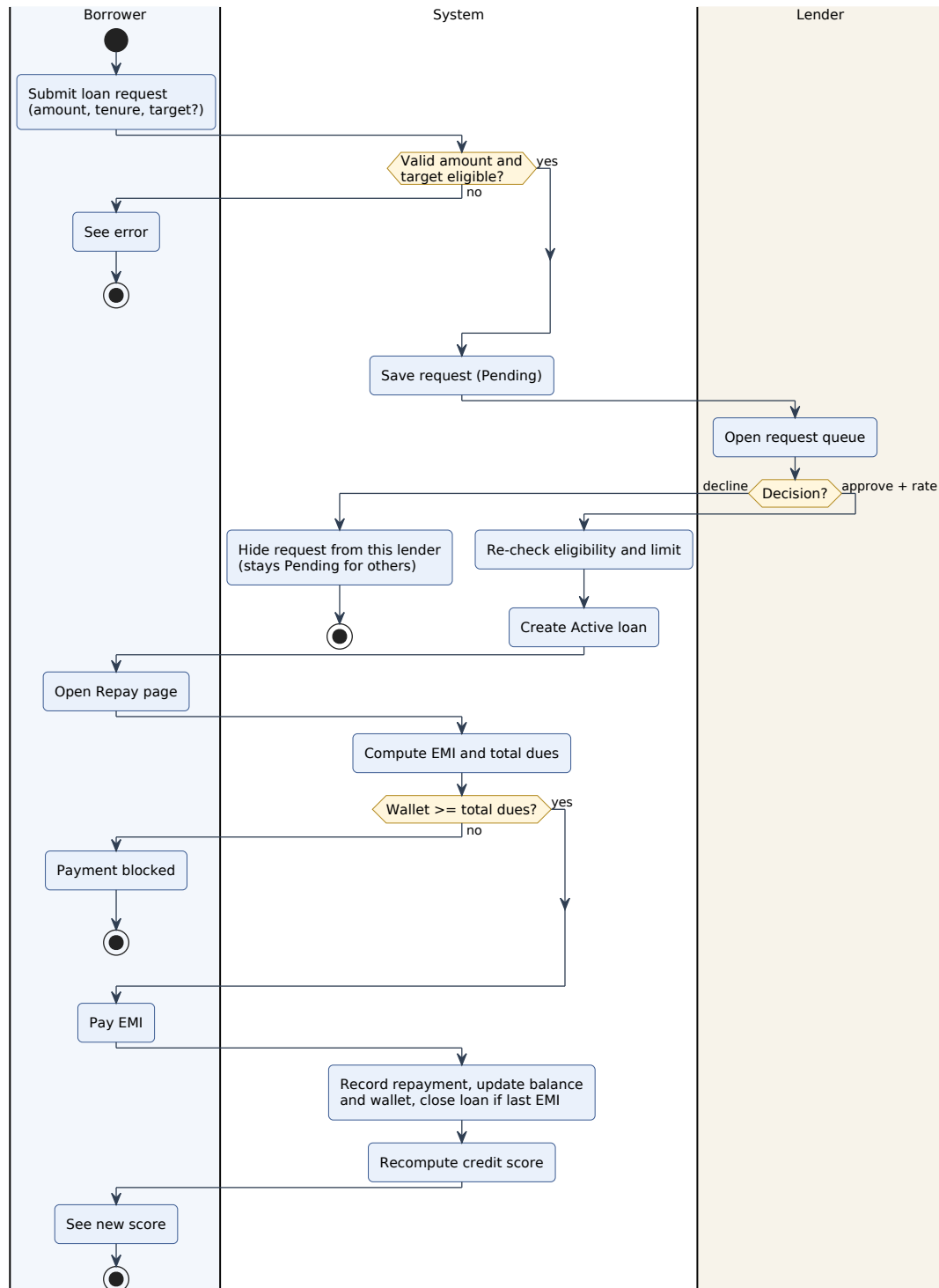


Figure 9: Activity (swimlane) diagram: loan request, approval and live repayment

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